



Society of Petroleum Engineers

SPE-227198-MS

Unlocking Hydrocarbon Potential: Petroleum Perspectivity Evaluation of the Jurassic Petroleum System, North Ustyurt Basin, Kazakhstan

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227198-MS](https://doi.org/10.2118/227198-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Objectives/Scope: The main objective of this study was to develop and analyze a basin and petroleum systems model (BPSM) to investigate potential reasons for previous unsuccessful exploration campaigns in the North Ustyurt Basin, an intracratonic basin with Mesozoic-Cenozoic infill of West Kazakhstan. With this 2D BPSM we aim to improve our understanding of the Jurassic petroleum system to evaluate its hydrocarbon potential and reduce exploration risks

Methods, Procedures, Process: Thermal history calibrated to vitrinite reflectance was reconstructed using 1D BPSM to assess timing of expulsion. Hydrocarbon charge was simulated in quick look 2D BPSM, while risk assessment accomplished with a commercial exploration risk, resource, and value assessment software, generating multiple Monte-Carlo simulations. The following petroleum system components and processes were assessed: trap, seal, source, reservoir and migration. Reservoir and fluid properties were obtained from interpretation of well data, while seismically derived maps were used to assess trap geometry. Timing of trap formation and charge as outputs of paleo reconstruction with Geological Chance of Success (GCOS) maps were used to evaluate and rank identified prospects.

Results, Observations, Conclusions: The principal source rocks of the region are enclosed within the Early-Middle Jurassic, showing the highest organic richness and sufficient maturity levels at the most subsided central part of the basin. According to BPSM results, Early Jurassic source rocks have reached the wet gas to gas window, while Middle Jurassic rocks reside in the main oil window. Significant burial during the Early Cretaceous resulted in hydrocarbon expulsion from the two Jurassic source rocks and also improved the sealing efficiency of the Jurassic cap rocks indicating favorable timing for the petroleum systems. This was taken as the main criterion to assess the charge probability when evaluating and risking the prospects.

Five main prospects within the basin surrounding the main kitchen area were identified. The kitchen and migration pathways were numerically simulated to assess the overall potential of each prospect accounting risk factors to calculate GCOS. Final resources vs. GCOS values summarize geological uncertainties

identified in this study. According to the outcome the main exploration risk in the region is related to availability of traps and the timing relationship of trap formation and hydrocarbon expulsion.

Novel/Additive Information: The geology and reservoir characteristics in combination with the distribution of potential source beds, timing of hydrocarbon generation, geochemical evidence for migration, and chances of preservation were used to rank the potential of Jurassic stratigraphic intervals in different parts of the basin. The results emphasize the potential of Jurassic source rocks and provide critical insights for further continued petroleum exploration in the North Ustyurt Basin.

Introduction

The purpose of the undertaken exploration study was to assess the petroleum system of the North Ustyurt Basin. The North Ustyurt Basin is located in the western part of Kazakhstan surrounded by the prolific Precaspian, Buzachi, and South Mangyshlak petroleum provinces. The basin potential remains poorly understood, with exploration drilling conducted long ago during the 1960s and 1970s. A limited number of hydrocarbon accumulations have been discovered in the basin since that time.

Regional geological surveys were accomplished during the Soviet era, with the purpose of evaluating the Jurassic, Triassic and Tertiary sequences. As a result, numerous prospects were detected and prepared for deep drilling, such as Gagarinskaya, Amanjol, Astaoi, Beyneu, Zhailgan, Terenskaya, Tereshkovskaya, Manashi, Sheluran, Tepke, Ashitaipak, Irdali, Kamennaya, Sarga, Tyshkandy, and Adamli (Ozdoyev, 1977). However, the results of exploration drilling during the Soviet era in these areas appeared pessimistic, showing dry tests, scarce occurrence of oil and gas shows and few commercial discoveries. Despite this, the recent discovery of commercial hydrocarbon accumulations in the Uzbekagaliyev field in 2021 provided new impetus for reevaluation of the oil and gas potential of the region. Pessimistic drilling results obtained during the earlier exploration campaigns could be related to the use of heavy barite-based mud during drilling, which could lead to the underestimation of the overall hydrocarbon potential (Radyushkina et al., 1989).

Both liquid and gaseous hydrocarbon accumulations are known to occur in the basin. Gaseous hydrocarbon accumulations are detected in the Eocene sandstones enclosed within the northern part of the basin at shallow depth. Liquid hydrocarbons are reservoired in the western edges of the basin within the Jurassic section. The hydrocarbons are believed to be sourced from the Jurassic continental lacustrine and coaly beds (Ulmishek, 2001) with some additional deep basinal sources in Paleozoic existing in the eastern part of the basin (Ulmishek, 2001).

The focus of the present study was the reevaluation of petroleum systems of the basin based on comprehensive core, log and seismic interpretation analyses integrated into basin modeling simulation.

Geological setting

The North Ustyurt Basin is situated in the northern part of the Ustyurt Plateau, occupying territory of Kazakhstan and Uzbekistan. Small parts of the basin extend into the Caspian Sea and Aral Lake (Figure 1). The northwestern boundary of the basin is traceable along the southern flank of the Paleozoic South Emba.

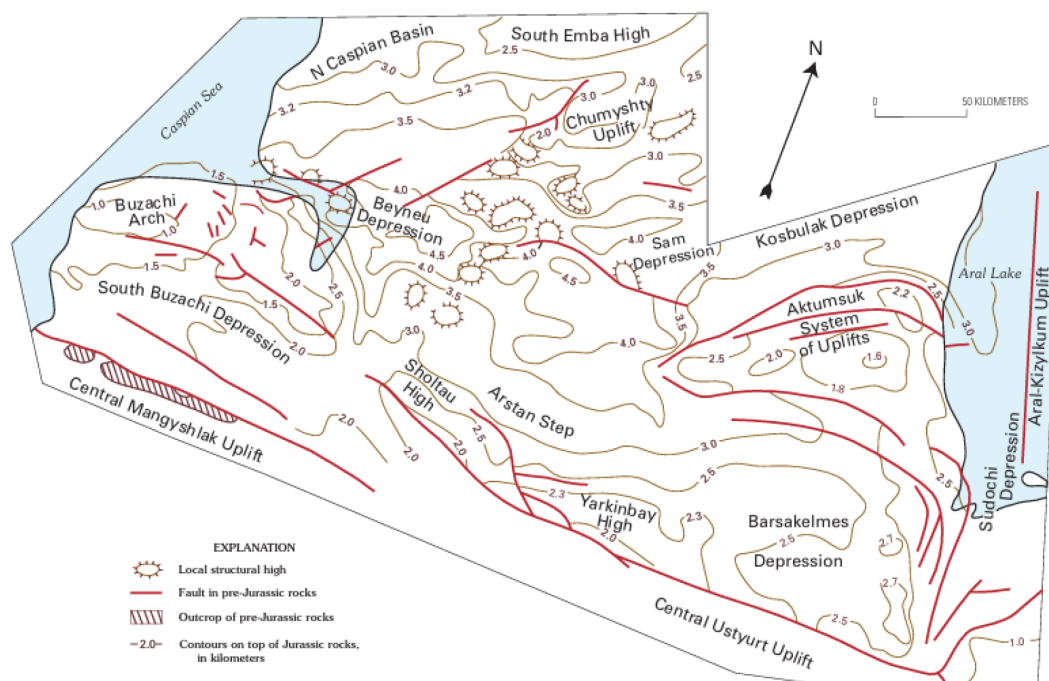


Figure 1—Structural elements within the North Ustyurt (Ulmishek, 2001).

A brief and concise description of the tectonic history of the North Ustyurt Basin is provided in [Ulmishek \(2001\)](#). Below is a summary of key elements.

The North Ustyurt Basin is filled with Jurassic-Tertiary sediments, unconformably overlying pre-Jurassic intermediate complex, structurally deformed and poorly understood. The basement is stratigraphically heterogeneous ([Babadzhanov et al., 1986](#)) and its age is not well known yet. Some estimates suggest Early Proterozoic, akin to Russian craton ([Khain, 1977](#)), Late Proterozoic ([Milanovsky, 1987](#)), and Late Paleozoic Hercynian ([Letavin, 1980](#)).

In the northern part of the basin within the Beyneu, Sam, and Kosbulak depressions the basement resides at deeper depths of 9 to 11 km, possibly consisting of oceanic or thinned transitional crust. In the southern zone of the uplifts, the basement at depths of approximately 5.5 to 8 km (Sholtau and Yarkinbay highs, Arystan step, and Barsakelmes depression) forms a stable block of granite crust of possible Precambrian age ([Babadzhanov et al., 1986](#)). A representative geologic cross-section through the western part of the North Ustyurt Basin and adjacent South Emba monocline, reflecting basin architecture is shown in [Figure 2](#).

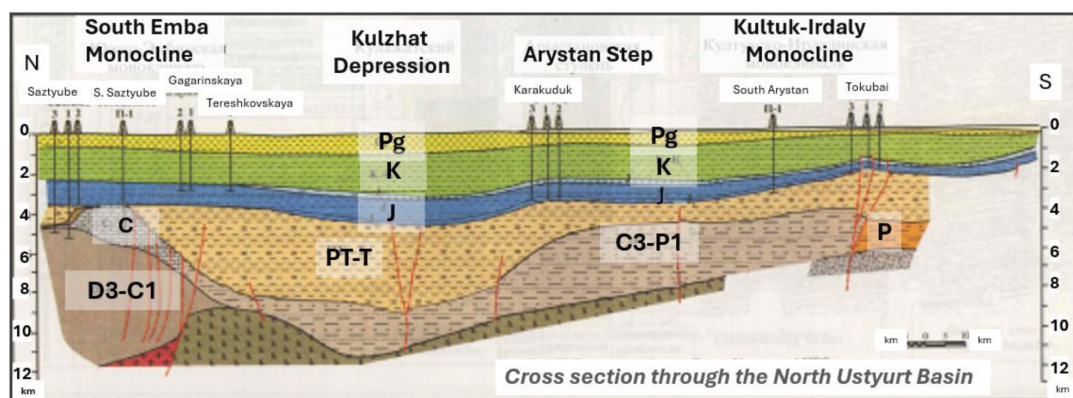


Figure 2—Schematic representation of cross-section through the North Ustyurt Basin and adjacent South Emba monocline (modified from [Daukeyev et al., 2002](#)).

During the Late Devonian to Early Permian time, the North Ustyurt Basin formed as a cratonic fragment, composed of platform-type carbonates and clastics, unconformably lying on top of older rocks.

North Ustyurt block collided with Russian craton along the South Emba suture, however the timing of this event is not well understood yet. Some probable estimates include pre-late Visean and Early Permian (Artinskian). The North Ustyurt underwent collision along the southern suture zone during Early Permian. On the eastern side, the basin collided with the Kazakhstan continent and the Russian craton with development of suture zone of Hercynian age (Late Carboniferous-Early Permian). Hercynian deformations are deeply buried and poorly known. Development of uplifts in the North Ustyurt due to Hercynian events that was subject to erosion can be inferred from the presence of thick Artinskian clinoforms north of the South Emba high (Sapozhnikov et al., 1986) consisting of clastics eroded from the North Ustyurt.

During the Late Permian–Triassic period, the basin was influenced by a rifting process in the area of the present Central Mangyshlak uplift along the southern border of the North Ustyurt block with north-south extension. The sediment supply, however, was located in the east, where orogenic movements continued.

In the later Triassic period or earliest Jurassic period, the closing of Paleo-Tethys in the south and collision of the Iran and other continental blocks with the Eurasian Tethyan margin occurred. The rift developed in the Central Mangyshlak was compressed and inverted, with thrusting in some areas of the North Ustyurt. The most intensive thrusting occurred in the Buzachi Peninsula and offshore and on the eastern side of the region.

Starting from the Jurassic period, the North Ustyurt was subject to subsidence with accumulation of 5 km of sediments. Only soft deformations affected the basin, some of which were related to rejuvenation of Triassic thrust movements. Present-day Ustyurt Plateau formed in post-Sarmatian time due to uplifting. The Plateau is covered by Sarmatian (upper Miocene) limestones and sandstones.

A description of generalized lithostratigraphy of the North Ustyurt Basin can be found in Daukeyev et al. (2002) (Figure 3). The most well studied sedimentary rocks in the basin are Jurassic-Paleogene in age. Triassic period rocks are studied in less detail, while little is known about older rocks. The sedimentary sequence is generally subdivided into pre-Triassic, Triassic, Jurassic-Cretaceous and Paleogene sections. Below is a short lithostratigraphic description.

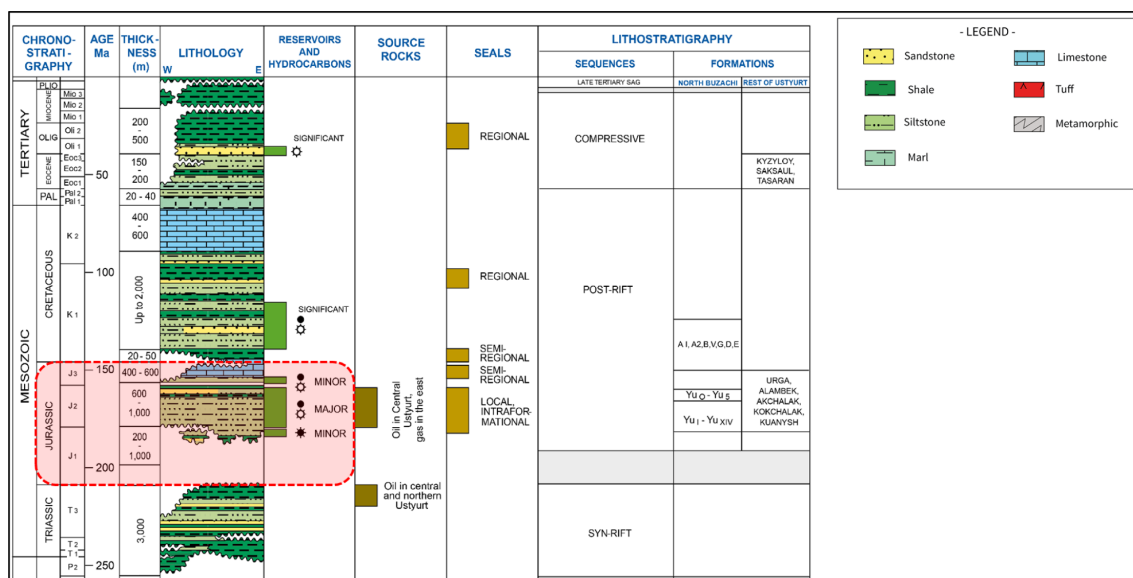


Figure 3—Generalized stratigraphy of North Ustyurt Basin.

Jurassic sedimentary rocks in the North Ustyurt Basin are predominantly siliciclastic, with the exception of the Upper Jurassic, where carbonates are present within the Oxfordian-Tithonian part of the sequence.

A high degree of carbonatization of sandstones decreases the reservoir properties. Clayey rocks in the North Ustyurt are also enriched with carbonate material. This is especially well expressed in Aalenian and Bathonian sediments, Callovian are less abundant in carbonate material.

Methodology

Understanding of Basin evolution with structural, stratigraphic, and subsidence frameworks and trends was accomplished through integration of observed well and seismic data with regional geologic background, inferred from published resources, including previous field studies and reserves estimation reports.

All existing geological and geophysical data over the northern, western and central parts of the North Ustyurt region have been collected, digitized and loaded into a unified database for further analysis and interpretation. In total, data from 20 wells, covering the entire study area, have been used in the present study. The data utilized for geological analyses and petroleum system evaluation included well logs, lithological descriptions of core and cuttings, mud gas logging, well test results (including drill stem test) and 2D regional seismic profiles and 3D seismic data.

Structural architecture of the basin was constrained with the use of mappable key horizons tied to specific geologic events. The key boundaries were first identified on well log data and further on seismic data, and then traced over the entire study area. The well-to-seismic tie procedure involved the generation of synthetic seismograms derived from vertical seismic profiling, sonic, and density logs to accurately define the time of the target horizons within the time seismic section. The studied sequence was further subject to detailed litho-stratigraphic well correlation to facilitate interpretation of depositional units and sequence stratigraphic events.

Laboratory geochemical analyses of bitumen extracts and mud gas logging data have been used to investigate the sedimentary section for organic geochemical characteristics in relation to the possible sources of hydrocarbons and migration pathways. Organic richness of the sediments was characterized according to published total organic content (TOC) measurements available in [Daukeyev et al. \(2002\)](#).

Thermal maturity of sedimentary organic matter was assessed using vitrinite reflectance values measurements, available only for one well, drilled in the central part of the region. For this reason, calibrated thermal history reconstruction was executed using 1D basin modeling, while hydrocarbon generation and migration (charge) processes were simulated in 2D basin modeling software. Multiple Monte-Carlo simulations were run with the use of commercial exploration risk, resource, and value assessment software. The following petroleum system components and processes were assessed: trap, seal, source, reservoir, and migration. Reservoir and fluid properties were obtained from interpretation of well data, while seismically derived maps were used to assess trap geometry. Thermal history calibrated to vitrinite reflectance allowed the assessment of timing of hydrocarbon expulsion, combined with geochemical proxies used for identifying source areas and likely migration pathways. Timing of trap formation and charge as outputs of paleo reconstruction with GCOS maps enabled evaluating and ranking of each prospect.

Results and Discussion

Geochemical evaluation

Exploration drilling results obtained during the Soviet era in the North Ustyurt Basin have been carefully reexamined to capture any even scarce indication of oil and gas presence according to the description of core samples, gas logging and well testing. Barite-weighted drilling mud aggressively used during drilling in the area was suspected to cause solids sag and formation damage by plugging pores leading to pessimistic results of the well tests.

Most of the wells drilled during early exploration campaigns tested dry, with few showing presence of dissolved gas and oil filament in water or drill mud. Overall, structural highs along the northeastern and

south central parts of the basin showed more encouraging results as opposed to the central sag part. While the western part of the region holds the main commercial liquid hydrocarbon discoveries, the northern part of the basin appears to be promising according to the occurrence of oil and gas shows and gas logging. Elevation of gas content according to gas logging is accompanied by an increase of wetness gas ratio. Some intervals in the northeast appeared as gas-bearing with methane content reaching 69.0 to 72.3%, with ethane in the range of 0.60 to 0.93%. Figure 4 shows typical relative abundance distribution of gaseous hydrocarbons in the northeast in comparison with western commercial oil and gas field. As can be seen in the histograms, the relative abundances of gaseous hydrocarbons are quite similar.

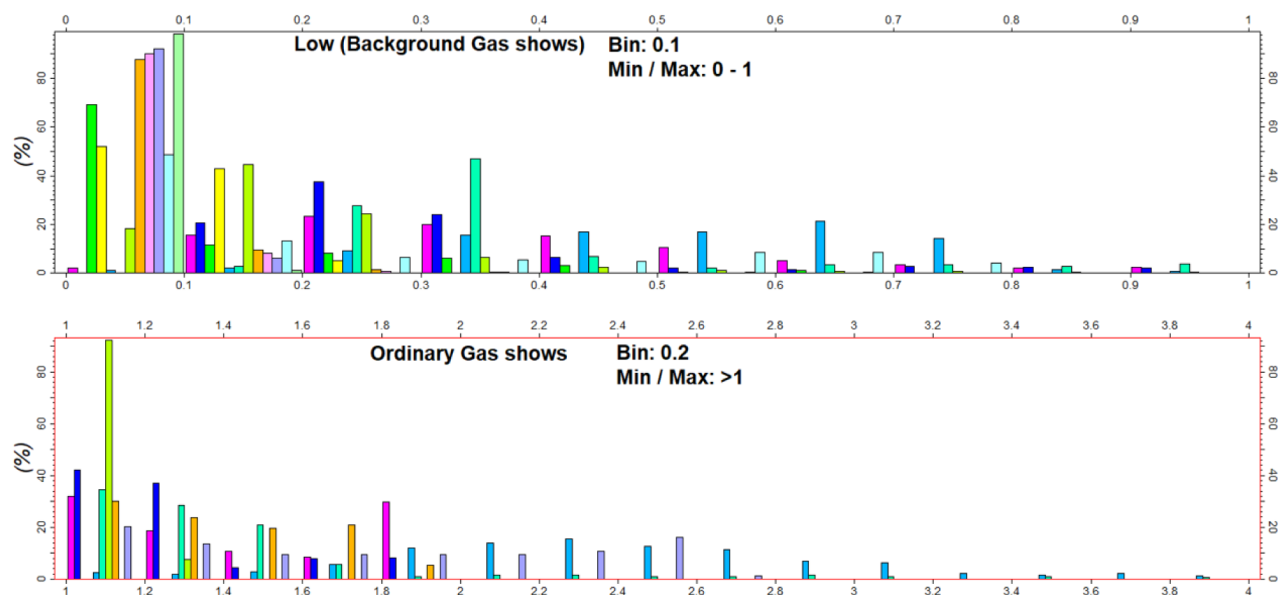


Figure 4—Histograms of summed gaseous hydrocarbons contents. Different colors represent different areas within the North Ustyurt.

Bitumen extracts collected from the basin depocenter are enriched in polar fraction, exhibiting yellow luminescence and more frequently occur in the Lower Jurassic section, as opposed to the Middle and Upper Jurassic. Similarly, gas logging anomalies tend to be more common in Lower Jurassic sediments. Dominance of saturated components are observable at the adjacent structural highs along with wet gas anomalies, oil and gas shows and, importantly, few existing oil discoveries. The lateral compositional changes from saturates dominated to more polar composition could be indicative of petroleum migration from the depocenter toward the uplifted edges. This presented observation is in agreement with the source rocks concept presented in previous studies (Radyushkina et al., 1989; Ulmishek, 2001).

Source rocks and thermal maturity. The principal source rocks in the North Ustyurt Basin are considered to be enclosed within the Early-Middle Jurassic sequence, characterized by the highest organic richness and sufficient maturity levels at the most subsided central part of the basin. Reports of organic content richness of sedimentary rocks in the region encompassing Paleozoic to Mesozoic sediments are reported in the published literature (Ozdoev, 1977, Daukeyev et al., 2002). The TOC values investigated sedimentary units achieve 1 to 2%, that is sufficiently high in terms of petroleum generation capacity. In some coal-rich samples, TOC values exceed 2% (Ozdoev, 1977, Daukeyev et al., 2002). The maturity levels of the source rocks fall into the oil window (0.7 to 1.2% Ro), implying that it has reached the stage of liquid hydrocarbon generation.

The source rocks contain predominantly continental type III kerogen, with abundant coalified plant remnants and occasional presence of coal. Pyrolysis measurements plotted on a Van Krevelen diagram fall mainly into a zone of mixed type II and III kerogen, showing comparatively lower HI values in the

range of 50 to 260 mg/g (Figure 5), in combination with lower OI dominantly not exceeding 30-40 mg/g. Predominance of type III kerogen in the Early-Jurassic succession is related to fluvial-deltaic and lacustrine depositional conditions (Pankina et al., 1981). Indicators of continental origin observed in core samples beside coal and abundant plant remnants include frequent crossbedding, fining upward, and development of scouring. According to regional paleogeographic studies (Pankina et al., 1981) the depositional environment in the basin took place in continental environment, ranging from lacustrine to fluvial-deltaic to the west and marine-lacustrine of the Caspian Sea to the southeast.

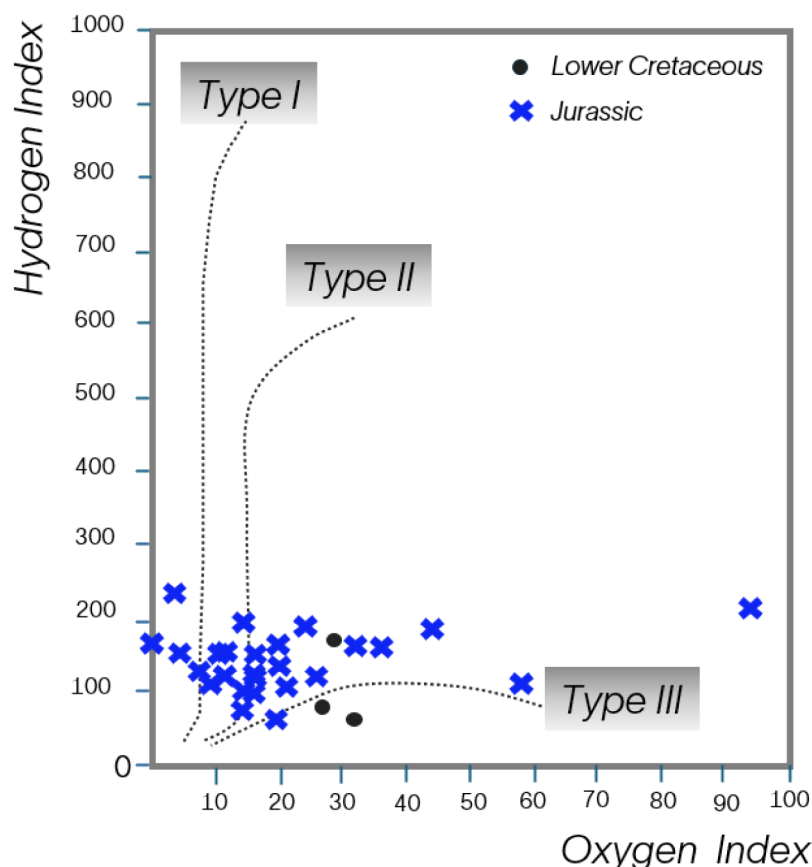


Figure 5—Source rock samples from the North Ustyurt plotted on a Van Krevelen diagram (reproduced from Daukeyev et al., 2002)

The underlying Permian-Triassic continental red beds are organically barren due to oxic depositional conditions unfavorable for the preservation of organic matter. The overlying Upper Jurassic and younger sedimentary units are organically poor and thermally immature, showing vitrinite reflectance values below the oil window.

Regional seal and reservoir rocks. Another crucial factor, restricting the overall perspective of overlying Cretaceous and younger rocks is the presence of a regional seal in the Upper Jurassic, preventing secondary migration of generated hydrocarbons from the underlying source rock units. The Upper Jurassic sequence consists of Callovian-Oxfordian marine shales and Kimmeridgian-Tithonian limestones, sealing the hydrocarbons entrapped in the underlying Jurassic sequences. The known commercial liquid hydrocarbon accumulations in the study area are reservoirized in the Early-Middle to Upper Jurassic Callovian Oxfordian rocks.

The reservoir rocks are of fluvial-deltaic origin, as indicated by regional paleogeographic studies reported by Pankina et al. (1981). The North Ustyurt constituted part of the large gently plunging in the south-eastern direction fluvial-deltaic plain of the Precaspian-Turanian region, fed by several large river systems, such

as Paleo-Volga, Paleo-Ural and Paleo-Emba. The highest coarse-grained sediment supply was distributed by the channel system during the Early Jurassic, when a high-energy environment prevailed due to abrupt topographic gradients. This is reflected in the development of basal sediments at the base of the Early Jurassic and Aalenian due to the influence of active scouring by alluvial processes. The Early Jurassic sequence is interpretable as an amalgamated braided river sequence, with maximum thickness reaching up to 300 m, closer to the depocenter part of the basin (Figure 6). Toward the basin uplifted margins, the Early Jurassic gradually pinches out from 50 to 100 m to a complete absence (Daukeyev et al., 2002).

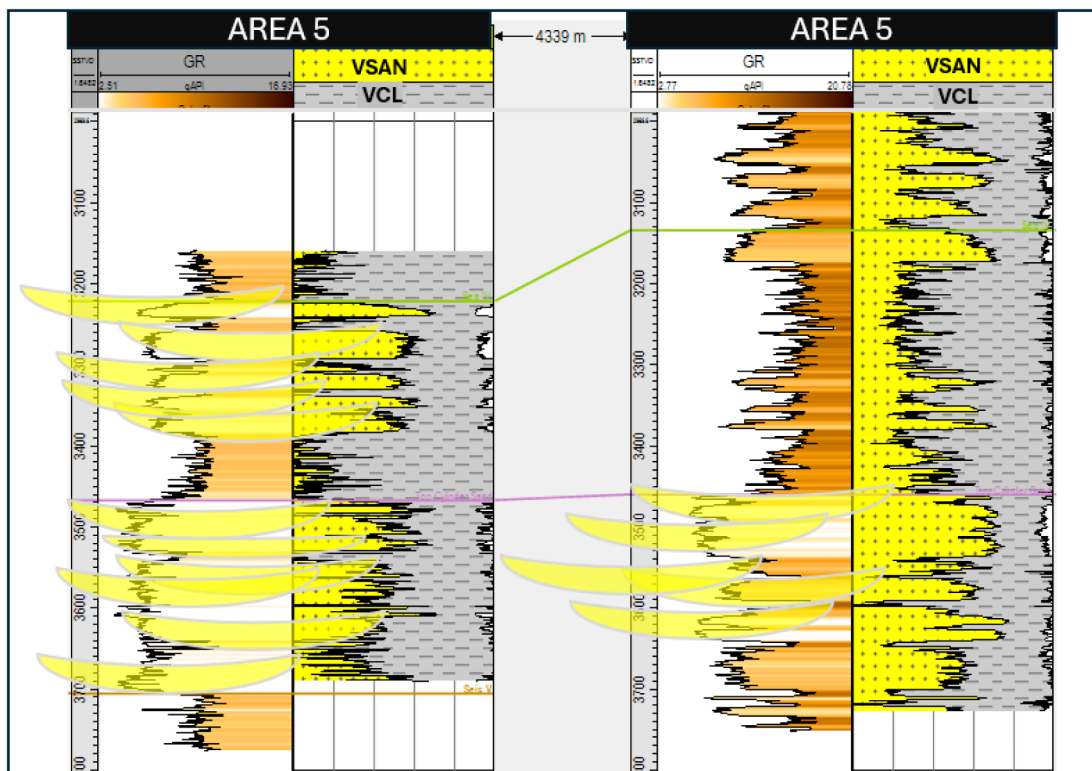


Figure 6—Detection of cylinder-shaped channelized sequences in the study area.

By the end of the Aalenian, the coarse-grained deposition diminished replaced by low gradient meandering, lacustrine to marshy environment, buried under Upper Jurassic marine transgressive sediments. The Early Jurassic rocks, although diminished in thickness in the uplifted marginal parts of the basin, are well developed as thick amalgamated alluvial beds closer to the subsided basinal parts to be charged by hydrocarbons generated from the surrounding source beds of the same age from the kitchen areas.

Thermal history: 1 D modeling. Thermal history calibrated to vitrinite reflectance measurements enabled assessment of source rock maturities and timing of hydrocarbon expulsion. Limited vitrinite reflectance measurements available for one well at the depth of 1625 –to 2400 m (Figure 7), drilled in the northwestern part of the basin, have been compared to the previously reported estimates published in Daukeyev et al. (2002) (Figure 8). A good match was found for the sample at the depth of 2500 m, corresponding to the vitrinite reflectance of 0.8%, R_o , while at the depth of 1700 m the estimate of 0.7% appeared to be by 0.1% higher comparatively to the dataset in Daukeyev et al. (2002), represented by a larger number of samples and importantly covering deeper parts of the sedimentary section. For thermal evaluation, a present day heat flow value of 55 mW/m² and geothermal gradient of 33 °C/km were set according to published resources, as the other constraints for the calibration. To evaluate organic maturation in the deepest central part of the basin a pseudo-well ‘Kitchen-1’ was generated for simulation purposes.

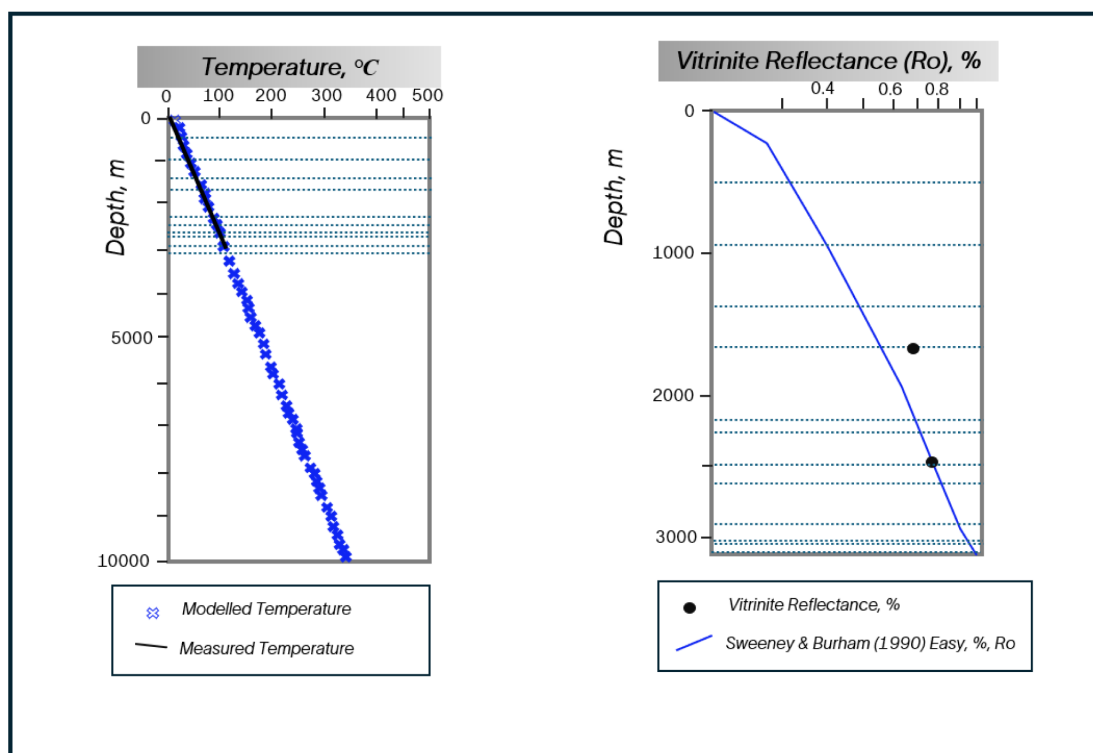


Figure 7—Thermal calibration to vitrinite reflectance values of a well drilled in the northwestern part of the basin. Vitrinite reflectance (right), temperature (left).

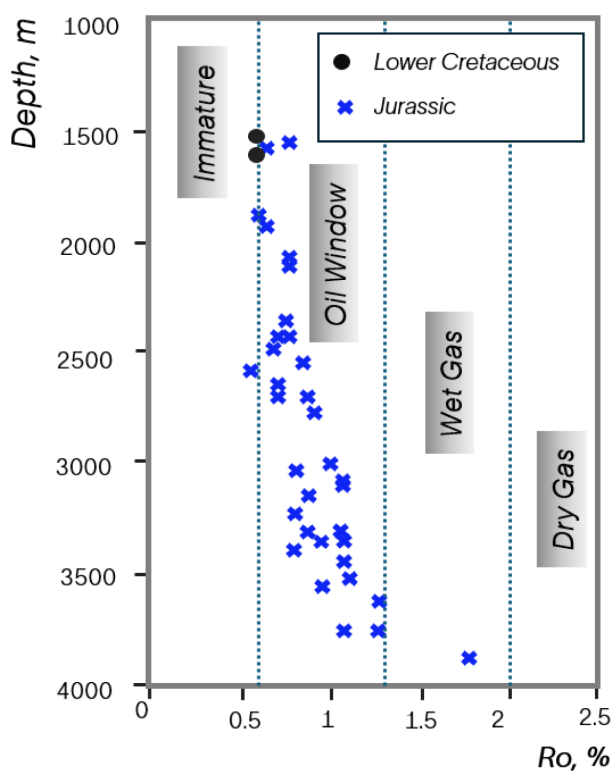


Figure 8—Vitrinite reflectance measurements vs. depth in the North Ustyurt Basin and interpreted maturity level (reproduced from Daukeyev et al., 2002).

The results of thermal evaluation for the pseudo well with vitrinite reflectance and temperature values and corresponding burial history are illustrated in Figure 9. The blue color in the left plot implies immature

conditions, dark green – early/waxy oil, medium green – the main stage of oil window, light green – the late oil window, dark red – wet gas, and light red – dry gas. The plot on the right of the figure depicts the Transformation Ratio (TR), defining the percentage of organic matter transformed into liquid hydrocarbons calculated for the source rock interval, for which appropriate kinetic conditions were set.

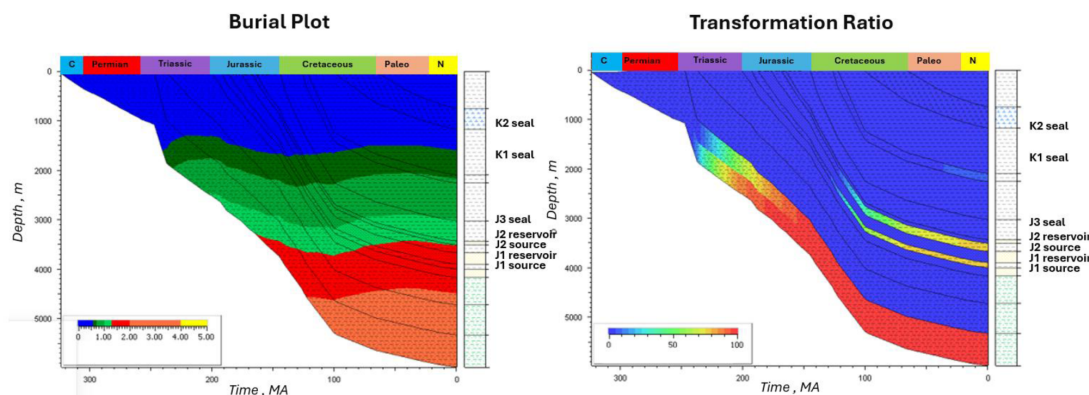


Figure 9—Burial (left) and TR (right) for pseudo-well Kitchen-1.

According to the BPSM results, Early Jurassic source rocks are within the gas window at present day, while Middle Jurassic source rocks reside in the main oil window. The trap formation in the Middle Jurassic according to the tectonic history of the basin was taking place due to inversion processes of the Central Mangyshlak rift system. The model predicts hydrocarbon accumulations in Jurassic traps triggered by burial during Cretaceous which:

1. improved sealing efficiency of the cap rocks and
2. resulted in continuous generation and expulsion of hydrocarbons from Jurassic source rocks (Figure 9).

Timing of trap formation and hydrocarbon charge were related according to results of paleo reconstruction and basin history. The output was integrated into the geologic event chart, summarizing formation of the key petroleum system elements, including source rocks, reservoir rocks, seal, and trap (Figure 10). The chart was used to evaluate how the petroleum system elements related in the timeframe favoring hydrocarbon generation, charge and accumulation. This was taken as the main criterion to assess the charge probability when evaluating and risking the prospects.

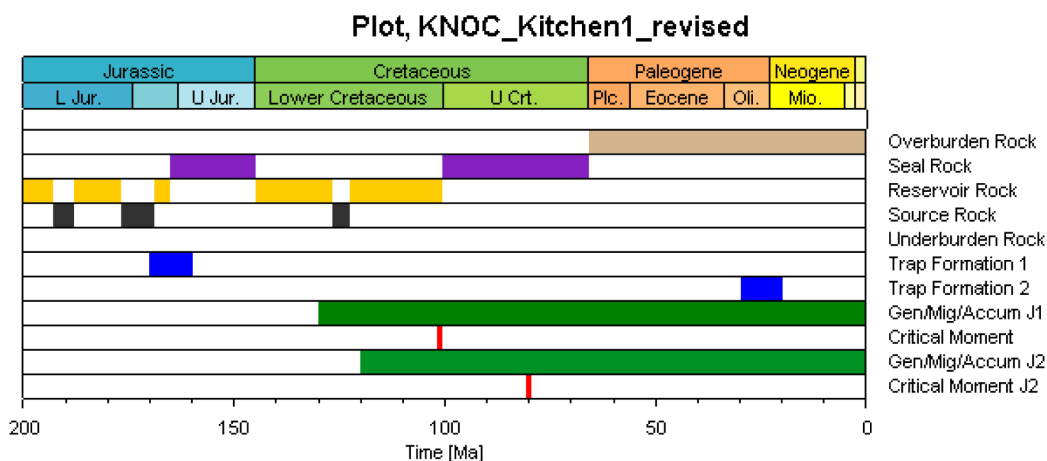


Figure 10—Geological events chart: Kitchen-1 pseudo-well (critical moment of the source rocks is plotted as 50% of TR).

2D Basin modeling

Hydrocarbon charging and accumulation simulation to predict hydrocarbon volumes was accomplished by map-based quick-look simulation software using the same input parameters applied in 1D BPSM of the Lower Jurassic and Middle Jurassic units. Figure 11 characterizes the obtained areal distributions of estimated Vitrinite Reflectance and TR values calculated for the source rocks. The results indicate advanced stage of thermal maturation of the Early Jurassic source rocks corresponding to the gas window buried in the depocenter of the basin. In the surrounding areas, the Early Jurassic source rocks reside at the late oil window. The overlying Middle Jurassic source rocks reached the main oil window over the main extent of the basin with the TR hardly reaching the critical moment (TR 50%) in the structurally elevated parts of the basin.

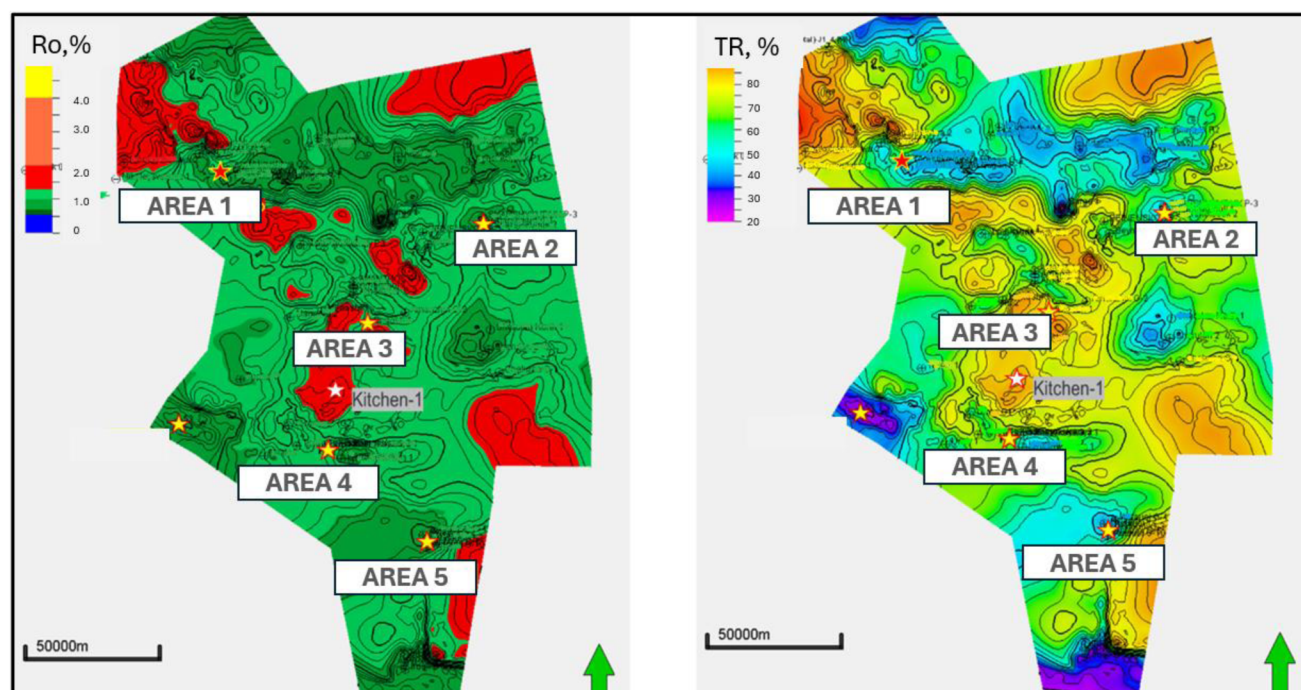


Figure 11—Geometrical parameters used as the input for simulation (northern, central and western parts of the North Ustyurt Basin).

At the same time the obtained results indicate a high chance of formation of accumulations with significant gas caps in the areas surrounding the highly mature zones (red zones). Hydrocarbon-charge modeling to simulate the migration pathways and formation of accumulations showed a significant amount of gas infilling the accumulations. This can be explained by the significant contribution of gas prone continental type III kerogen to the source rocks of the studied areas, for which the Pepper-Corvi type III kinetic model was applied.

Geologic chance of success

Basin modeling was accomplished with risk analyses using probabilistic approach by exploration risk, resource, and value assessment software, implying generation of multiple Monte-Carlo simulations scenarios. Risk analyses encompassed the following petroleum system components and processes: trap, seal, source rocks, reservoir, and hydrocarbon migration. Reservoir and fluid properties were obtained from interpretation.

Five main prospects within the basin part surrounding the main kitchen area were identified (Figure 11). The input defining the trap geometry of each evaluated prospect with respect to size of the area and depth

was defined according to the corresponding interpreted well-tied seismic surfaces. The structure maps were generated as a result of depth conversion using a layer-cake velocity model calibrated by well tops. The closure in each run was set by varying the position of the fluid contacts from the crest of the structure down to the spill point following a normal distribution. Figure 12 shows the structure surface used as the maximum closure scenario and the trap geometries calculated by the software. Table 1 provides an example of the geometrical parameters used as the input for the simulation.

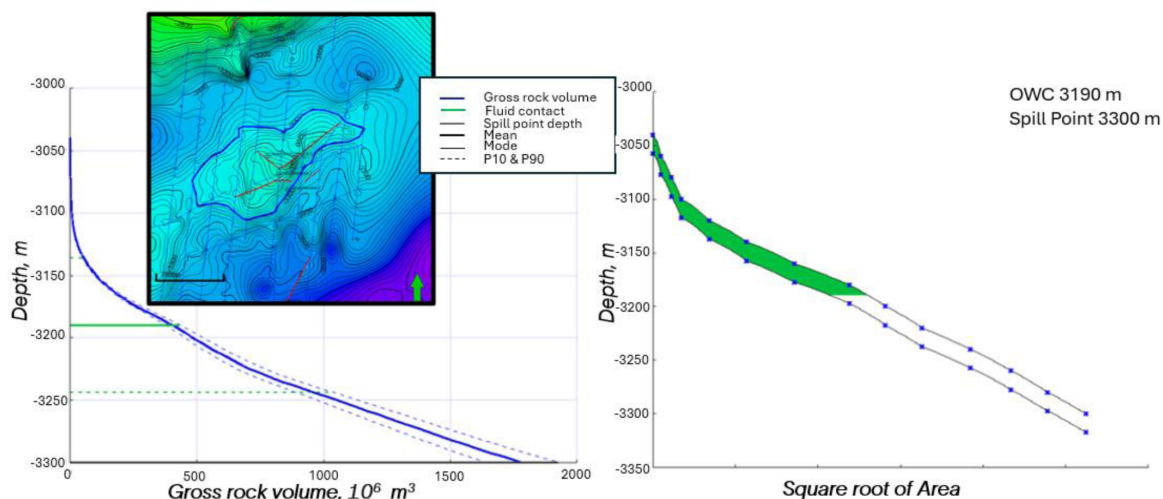


Figure 12—Example of trap geometry set in the Top Lower Jurassic.

Table 1—Example of geometrical parameters used in the evaluation.

		Area [km ²]				
Depth [m]	Dist. type	Mean	Std. dev.	Min (Lo)	Central	Max (Hi)
3040.0	Const	0.0	0.0		0.0	
3060.0	Const	0.0363	0.0		0.0363	
3080.0	Const	0.201	0.0		0.201	
3100.0	Const	0.483	0.0		0.483	
3120.0	Const	1.86	0.0		1.86	
3140.0	Const	5.15	0.0		5.15	
3160.0	Const	11.7	0.0		11.7	
3180.0	Const	22.8	0.0		22.8	
3200.0	Const	31.8	0.0		31.8	
3220.0	Const	42.6	0.0		42.6	
3240.0	Const	59.4	0.0		59.4	
3260.0	Const	75.3	0.0		75.3	
3280.0	Const	91.6	0.0		91.6	
3300.0	Const	110.2	0.0		110.2	

Parameter [Units]	Dist. type	Mean	Std. dev.	Min (Lo)	Central	Max (Hi)
HC water contact [m]	Norm	3190.0	41.1	3080.0		3300.0
Spill point depth [m]	Const	3300.0	0.0		3300.0	
Reservoir thickness [m]	Unif	17.5	1.1	15.6		19.4

Net pay, porosity, and oil saturation applied were estimated based on log data. For the prospects with one well available, a constant value of net pay was applied, while for the prospects with more than one well, the range was set to follow uniform distribution between minimum and maximum. According to the estimates obtained, the net pay in the prospects falls in the range of 2 to 98 m. The range of porosity and oil saturation was set according to calculated probability distribution functions. The maximum and minimum values of

porosity and oil saturation correspond to 0.07 to 0.25 and 0.41 to 0.86, respectively. The formation volume factor for oil (B_o) was set to equal 1.36 and the recovery factor to 0.3, in accordance with field data.

GCOS was calculated for each prospect accounting for five risk components: trap, seal, source rocks, reservoir and hydrocarbon migration. A constant chance of success equal to 0.8 was applied to seal for all prospects, considering documented ubiquitous occurrence of regional tight carbonates in the Upper Jurassic and numerous continuous intermittent shaly intervals in the Middle Jurassic. The chance of success values applied for reservoir, trap and source rocks were varied in the range of 0.7 to 0.9, depending on the field observations. Comparatively higher values were applied in accordance with the actual presence of the examined elements, confirmed by analyses of logs, core and seismic data. The probability of migration was assigned considering the results of hydrocarbon charge simulation, together with timing of trap formation.

The main uncertain parameter to play a crucial role in the risk assessment was linked to the availability of traps by the onset of hydrocarbon expulsion. Assuming an optimistic scenario for trap formation, the overall results of ranking assessment indicate the high-grade areas are distributed in the western part of the basin, comprising around 80-90% of the entire estimated volumes. Higher risks can be encountered in the east of the central part of the basin, where net pay decreases.

Conclusion

This work provides an assessment of petroleum system in the North Ustyurt Basin, contributing crucial insights for future exploration. The Early-Middle Jurassic source rocks emerge as the most promising source rock within the region, exhibiting good organic matter richness and quality according to geochemical indices, lateral and vertical hydrocarbon concentration trends. All necessary elements of a petroleum system including source rocks, reservoir rocks, seal and traps are in place within the Jurassic sequence. Hydrocarbons were most probably generated in the organic-rich coal-bearing shales of the Early-Middle Jurassic in the subsided central part of the basin, migrating towards structurally uplifted margins of the basin via fault systems and associated fracture networks. According to the BPSM the Early-Middle Jurassic source rocks reached the critical moment ($TR=50\%$) in Late Cretaceous, after trap formation during Middle Jurassic. Early Jurassic multiple stacked braided river sandstones distributed up dip along migration pathways could represent the most favorable places to hold commercial accumulations. GCOS maps generated as the main output for evaluation and ranking of each prospect combined with geometrical proxies, showed the highest potential for the western part of the region. Assessment of uncertainties associated with petroleum system elements showed the highest risks to be associated with trap availability in place by the time of hydrocarbons expulsion.

Acknowledgement

Our gratitude is extended to SLB and KNOC for permission to publish the paper. Special thanks to Duplo Kornpihl, SLB Exploration Services Team Lead, Digital & Integration-DSS Solutions, Oliver Schenk, SLB Principal Petroleum System Geologist, Sabine Mehay, SLB Principal Geochemist, and Carlos Andres Sanchez Torres, SLB Digital Product Analyst for review of this paper and valuable insights. Special thanks to Dilyara Abdulmanova, KNOC Senior Geologist, for contribution in analyses and interpretations.

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